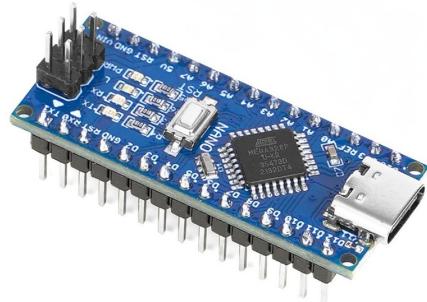


Lab 6: Serial input



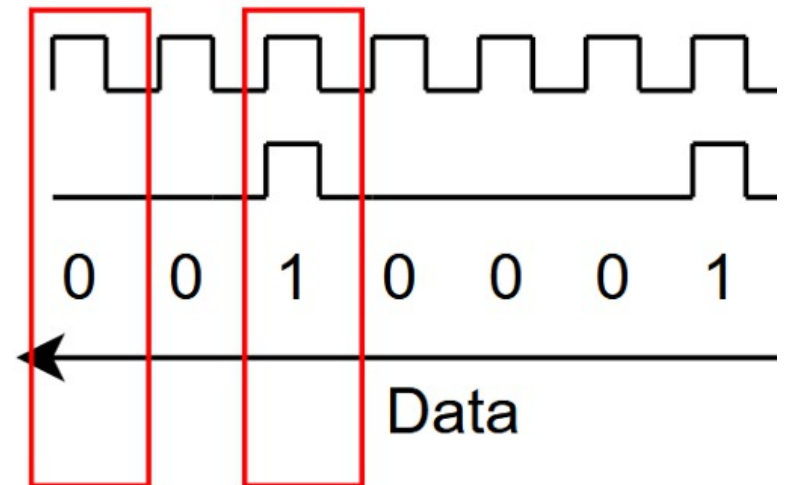
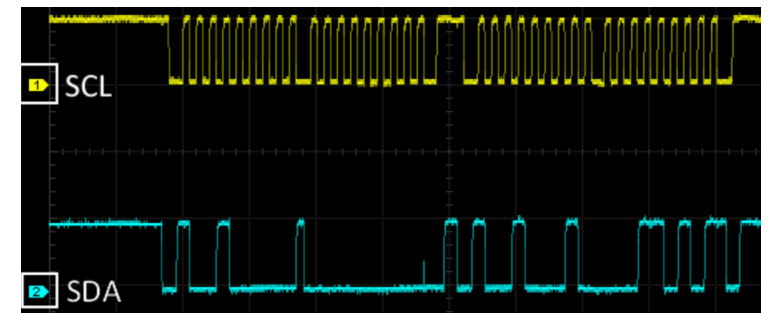
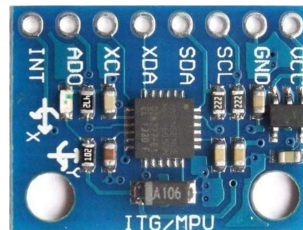
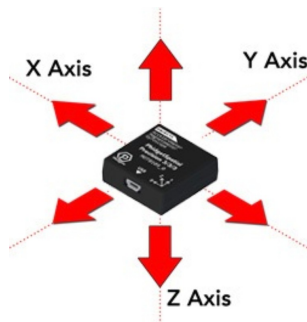
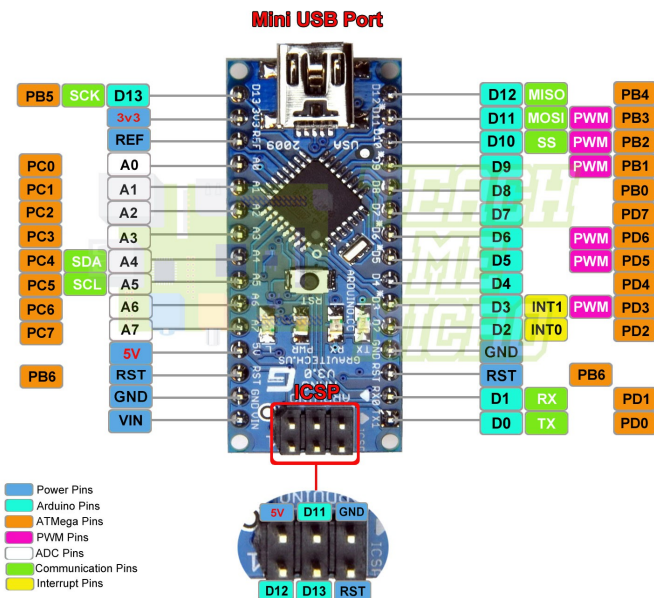
Lab goals

- Get acquainted with the microcontroller's capability for working with input devices via serial communications

Required hardware/software

- Everything from lab 5, plus:
- MPU-6050 sensor board x 1

ARDUINO NANO PINOUT



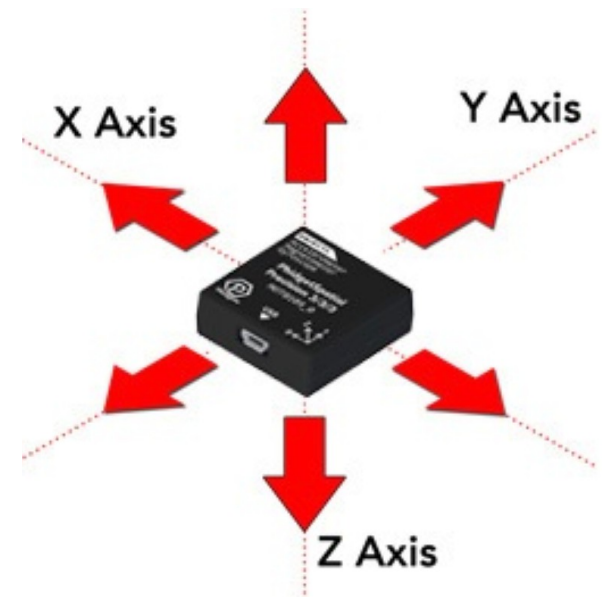
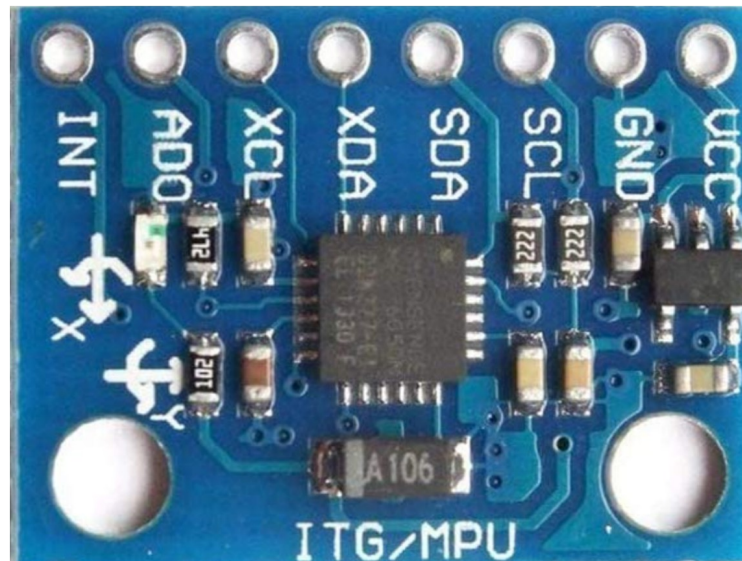
Lab 6: Serial input (background)

In this lab, we'll be working with a MPU-6050 development board.

The MPU-6050 is an Integrated Circuit (IC) (aka “chip”) that provides a 3-axis accelerometer and 3-axis gyroscope with an I²C interface, all in a single 4 mm x 4 mm package.

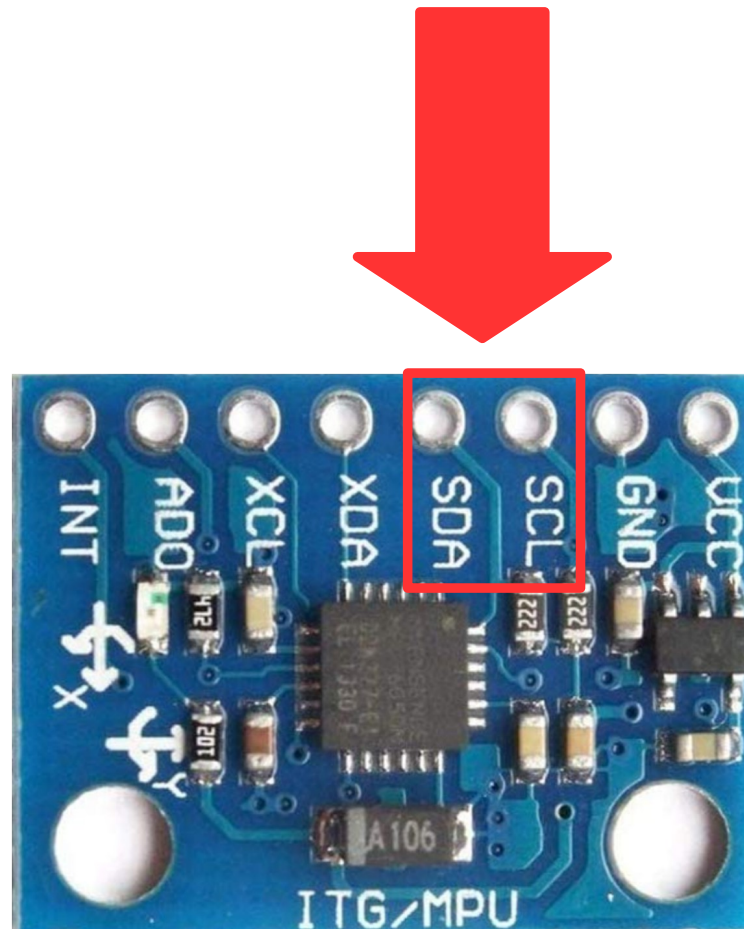
Sensors such as those offered by the MPU-6050 are, for example, how “smart phones” determine their orientation.

We will use the MPU-6050 sensor in lab 6 to add orientation as an input to our microcontroller.



Lab 6: Serial input (background)

Note the “SDA” (serial data) and “SCL” (serial clock) connections for I²C that we first saw in lab 5.

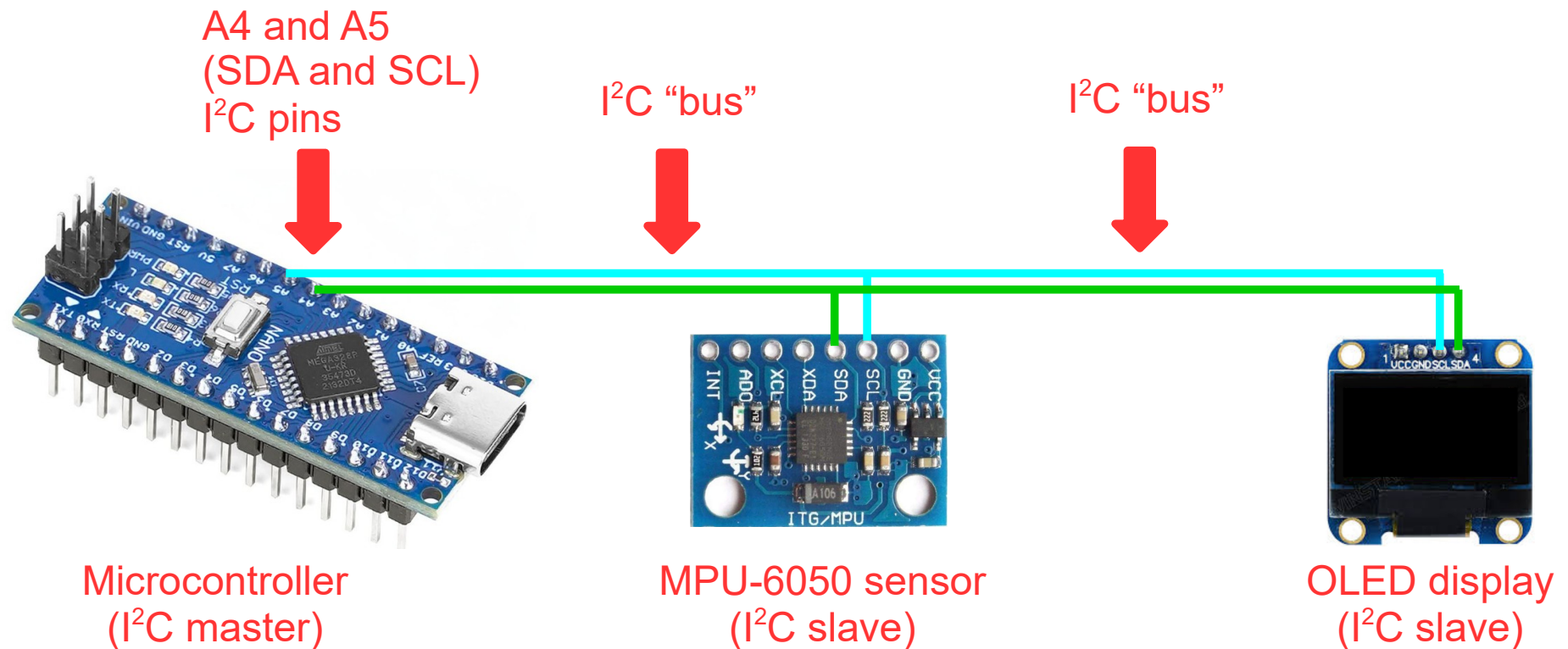


Lab 6: Serial input (background)

Serial input is not fundamentally different than serial output.

In practice, we do not tend to differentiate between “serial input” and “serial output” – we just work with “serial devices” – “I²C devices”, “SPI devices”, etc.

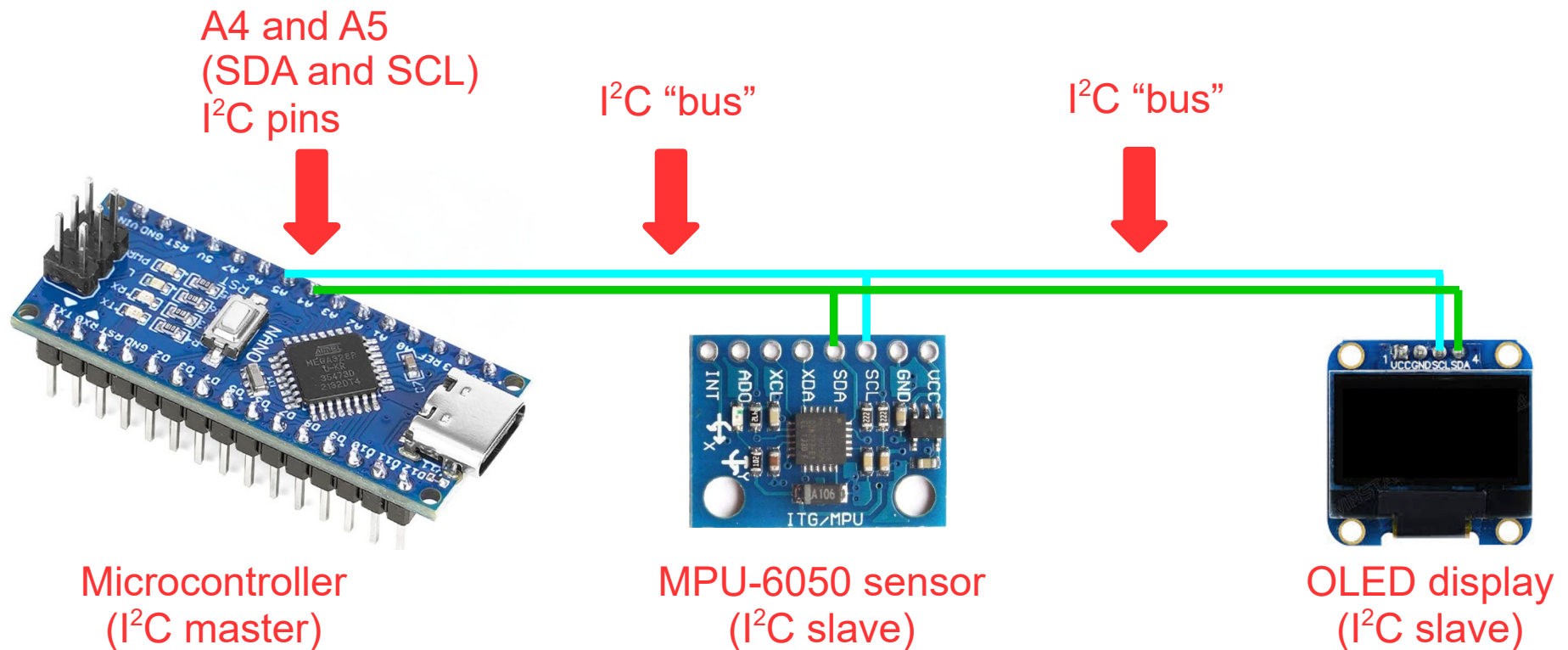
I have only really differentiated them here for the purposes of creating two distinct labs – one for serial “output” and the other for serial “input.”



Lab 6: Serial input (background)

Note that we now have two different I²C devices on the same “bus” (on the same set of wires).

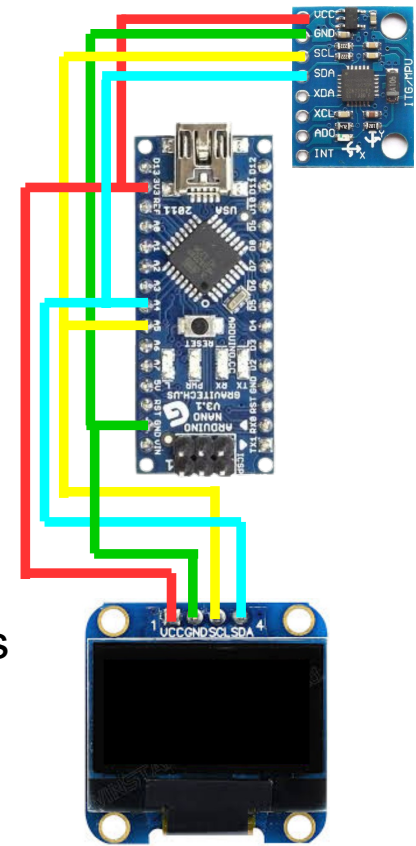
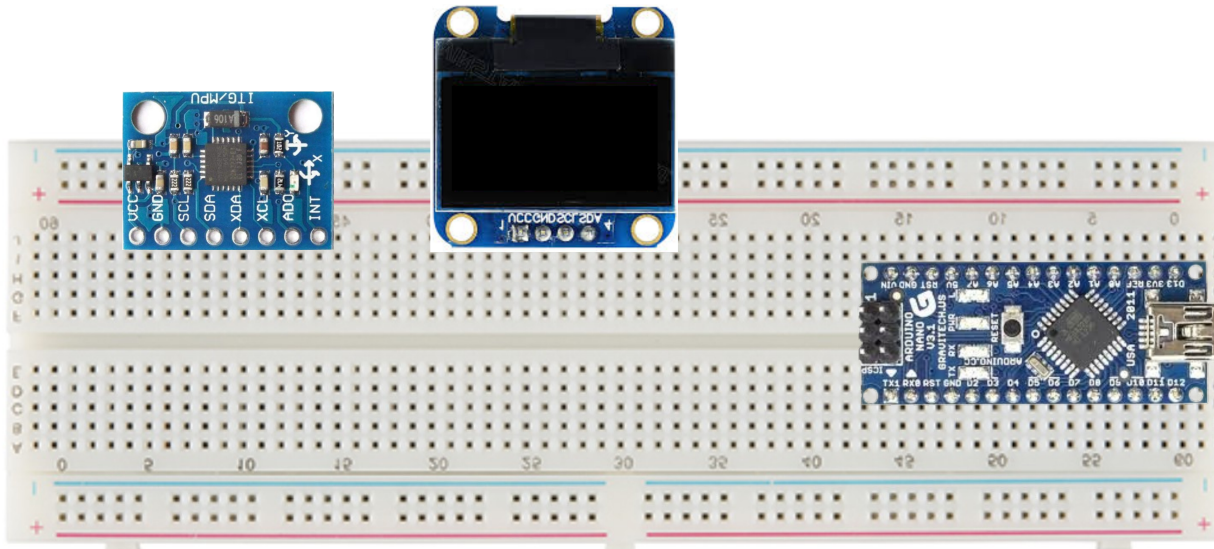
The program that we will be loading onto the microcontroller for lab 6 will continually ask the MPU-6050 for orientation information over the I²C bus and continually update the OLED display via the I²C bus using the latest orientation information. Both of these operations will happen about 20 times per second, or about once every 5 milliseconds.



Lab 6: Serial input

Steps

1. As a starting point, disconnect the USB cable from your development board and leave the components and connections from lab 5 intact.
2. Remove the potentiometer and associated wires from your breadboard.
3. Plug your OLED display into an outermost column of your breadboard as shown below.
4. Plug the MPU-6050 board similarly into your breadboard as show below.



(Next slide)

Lab 6: Serial input

Steps

5. Make the following connections between the OLED display, the MPU-6050 board, and your development board:

- VCC to 3V3
- GND to GND
- SDA to A4
- SCL to A5

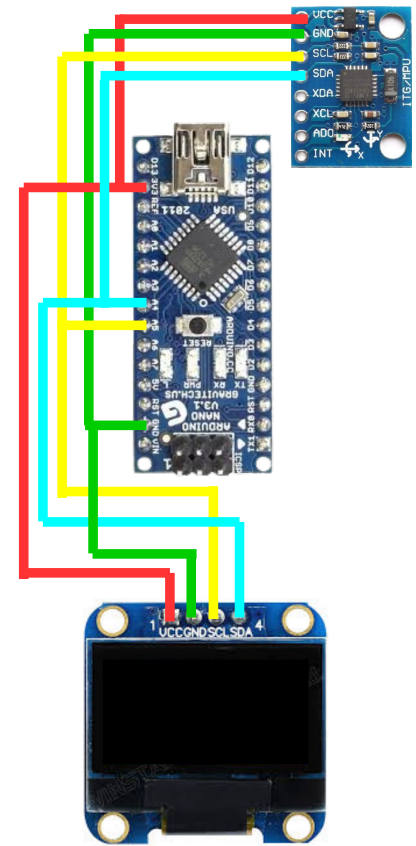
6. Download the ZIP file found, extract it into a directory named intro-microcontrollers-lab-006 in the same directory where all of your other Arduino projects are, and then open up its .ino file with Arduino IDE:

<https://github.com/dandandrea/intro-microcontrollers-lab-006/archive/refs/heads/main.zip>

7. Connect your development board to your laptop via the included USB cable.

8. Program your development board with the program displayed in Arduino IDE.

(Sketch > Upload)



(Next slide)

Lab 6: Serial input

Steps

9. Observe that as you tilt your breadboard (and thus the MPU-6050 sensor), the “cursor” displayed on your OLED moves in accordance.

NOTE: We are only using the X-axis and Y-axis acceleration readings from the MPU-6050 and are ignoring the Z-axis readings.

